Fiscal Year:	FY 2015	**Task Last Updated:**	FY 07/16/2015
PI Name:	Kassemi, Mohammad Ph.D.		
Project Title:	The Zero Boil-Off Tank (ZBOT) Experiment		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mohammad.kassemi@nasa.gov	**Fax:**	FY 216-433-5033
PI Organization Type:	NASA CENTER	**Phone:**	216-433-5031
Organization Name:	NASA Glenn Research Center/Case Western Reserve University		
PI Address 1:	NASA Glenn Research Center		
PI Address 2:	21000 Brookpark Road, MS110-3		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	10
Comments:	NOTE (Dec 2019): former affiliation included National Center for Space Exploration Research (NCSER), per information from J. McQuillen/GRC		
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	10/05/2007	**End Date:**	12/31/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
Contact Email:	john.b.mcquillen@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 12/31/2018 per J. McQuillen/GRC (Ed., 5/18/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chato, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	PI: Dr. Mohammad Kassemi (Case Western Reserve University, NCSER/NASA GRC); Co-I: Dr. David Chato (NASA GRC); Project Scientist: John Mcquillen (NASA GRC); Project Manager: Bill Sheredy (NASA GRC). NASA's projected exploration program includes a series of human and robotic expeditions to low and high Earth orbit, Moon, Mars, and possibly the asteroids and moons of other planets. Integral to all phases of these space and planetary expeditions is affordable and reliable cryogenic fluid storage for use in the propellant or life support systems. Without safe, efficient, and flexible cryogen storage, economically justified human missions may not be possible. Cryogen vaporization caused by various heat leaks into the tank from its surroundings is the main cause of storage tank self-pressurization. This has led to the development of innovative pressure control designs based on mixing of the bulk liquid with or without active cooling. Unfortunately, both tank pressurization and pressure control are governed by intricate and complicated dynamic interactions among the forced mixing, the various gravity-dependent or -independent transport mechanisms in the vapor and liquid phases, and the condensation/evaporation process at the interface. Consequently, effective implementation and optimization of a dynamic pressure control system for space applications cannot be accomplished solely by empiricism, especially, in view of serious scarcity of relevant microgravity data. Since large-scale in-space technology validation test or demonstration of the cryogenic storage tank prototype has not been possible due to both budgetary and time constraints, it was proposed that a series of small-scale targeted microgravity simulant fluid storage tank experiments can be performed aboard the International Space Station (ISS) to first understand the underlying scientific physical phenomena influencing tank pressurization/pressure control in space and then to validate and verify state-of-the-art two-phase computational fluid dynamics (CFD) models that can be used to scale-up the storage tank pressure control design for various in-orbit or on-surface applications. In this manner, the risks associated with the design of space-based storage tanks using only ground testing of the technology can be greatly reduced. In this light, the Zero-Boil-Off Tank (ZBOT) experiments have been developed as a series of small scale ISS experiments aimed at delineating important fluid flow, heat and mass transport, and phase change phenomena that affect cryogenic storage tank pressurization and pressure control in microgravity. The experiments use a simulant transparent low boiling point fluid (PnP) in a sealed transparent Dewar to study and quantify: (a) fluid flow and thermal stratification during pressurization; (b) mixing, thermal destratification, depressurization, and jet-ullage penetration during pressure control; (c) transport and kinetics effects of noncondensable pressurant gases on interfacial evaporation and condensation rates; and (d) transport, heat transfer, and phase change mass transfer interactions between evaporating and condensing sprayed droplets and ullage during active spray-bar-droplet cooling. The experiments will provide valuable microgravity empirical two-phase data associated with the above-mentioned physical phenomena through highly accurate local wall and fluid temperature and pressure measurements, nonintrusive Particle Imaging Velocimetry (PIV), and full-field phase-distribution and flow visualization. Moreover, the experiments are performed under tightly controlled and definable heat transfer boundary conditions to provide reliable high-fidelity data and precise input as required for validation & verification of state-of-the-art two-phase CFD models developed as part of this research and by other groups in the international scientific and cryogenic fluid management communities. During the project review process, the ZBOT Science Review Panel consisting of members from academia, NASA space & flight centers, and aerospace industry recommended that the broad scientific goals of the project can best be achieved through a series of consecutive experiments. Thus, the first experiment (ZBOT-1) in the series is focused on microgravity tank pressurization, thermal stratification, and destratification through mixing. The second experiment (ZBOT-2) aims at delineating the important effects of noncondensable pressurants on tank pressurization and pressure control. And the third experiment (ZBOT-3) investigates transport and phase change aspects of active tank pressure control by spray-droplet and forced-jet mixing/cooling. Implementation of the recommended strategy has resulted in a logical hierarchical series of three consecutive experiments whereupon the main tank, the fluid support unit, and the diagnostic rig are constructed during ZBOT-1 and augmented in a cost-effective step-by-step manner through implementation of additional units and diagnostics to accommodate the added complexities required by ZBOT-2 and ZBOT-3 experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The result of the ZBOT investigation will primarily and directly address the fundamental need of NASA for efficient, reliable, and enabling propellant and life support technologies for long-duration planetary exploration. However, the fundamental two-phase data and scientific understanding provided by the experiment and the associated theoretical model development will affect a diverse set of ground-based two-phase technologies such as condensing heat exchangers, phase separators, water-handling operations in fuel cells, and storage of cryogenic and non cryogenic consumable fluids.
Task Progress:	The Zero-Boil-Off Tank (ZBOT) Experiment is a small-scale experiment that uses a transparent ventless Dewar and a transparent simulant phase-change fluid to study sealed tank pressurization and pressure control with applications to on-surface and in-orbit storage of propellant cryogens. The experiment will be carried out under microgravity conditions aboard the International Space Station in the 2016 timeframe. This continued modeling effort presents preliminary results from ZBOT's ground-based research delineating both pressurization and pressure reduction trends in the sealed Dewar. Tank pressurization is accomplished through heating of the test cell wall in the wetted and un-wetted regions simultaneously or separately. Pressure control is established through mixing and destratification of the bulk liquid using a temperature controlled forced jet flow with different degrees of liquid jet subcooling. A Two-Dimensional axisymmetric two-phase CFD model for tank pressurization and pressure control is also presented. Numerical prediction of the model are compared to experimental 1g results to both validate the model and also indicate the effect of the residual non-condensable gas on evolution of pressure and temperature distributions in the ullage during pressurization and pressure control. Microgravity simulations case studies are also performed using the validated model to underscore the profound effect of the noncondensables on condensation rates and interfacial temperature distributions with serious implications for tank pressure control in reduced gravity. Additional modeling efforts have been undertaken regarding the velocity profile exiting the jet mixing nozzle and are in agreement with the data from the range of flow rates that were experimentally measured. The models then extended the range of confidence for slower flow rates that compromise the majority of the test matrix.

	The test matrix is being modified to maximize and accommodate the revision to the test plan regarding the flow visualization. Investigations have been initiated into other techniques to measure either flow or temperature profiles.
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